

MTH209: Assignment 7

The following code attempts to do a simulation to verify a central limit theorem for when

$$X_1, X_2, \dots, X_n \stackrel{iid}{\sim} \text{Gamma}(\alpha, 1).$$

The goal, in repeated simulations is to calculate and store

$$\sqrt{n}(\bar{X}_n - E(X_1))$$

and check whether it converges to $N(0, \text{Var}(X_1))$. The following are the settings:

1. $n = 50$
2. Replications = 100
3. $E(X) = \alpha$
4. $\text{Var}(X) = \alpha$.

There are logical and syntax errors in the code below. Upload a file that removes all the errors and returns the desired output.

```
## sqrt(n)(barx - E(x))
#
# alpha is the shape parameter of the gamma distribution
CLT_simulation <- function(alpha)
{
  reps <- 50
  n <- 1000

  store <- numeric(length = n)
  for(r in 1:reps)
  {
    sample <- rgamma(n, shape = 1/alpha)
    store <- sqrt(n)(sum(sample) - alpha)
  }
  return(store)
}
```

INSTRUCTIONS: Upload the file with the name `assignment7.R` on helloIITK. Make sure your code only has the function `CLT_simulation` and nothing else.